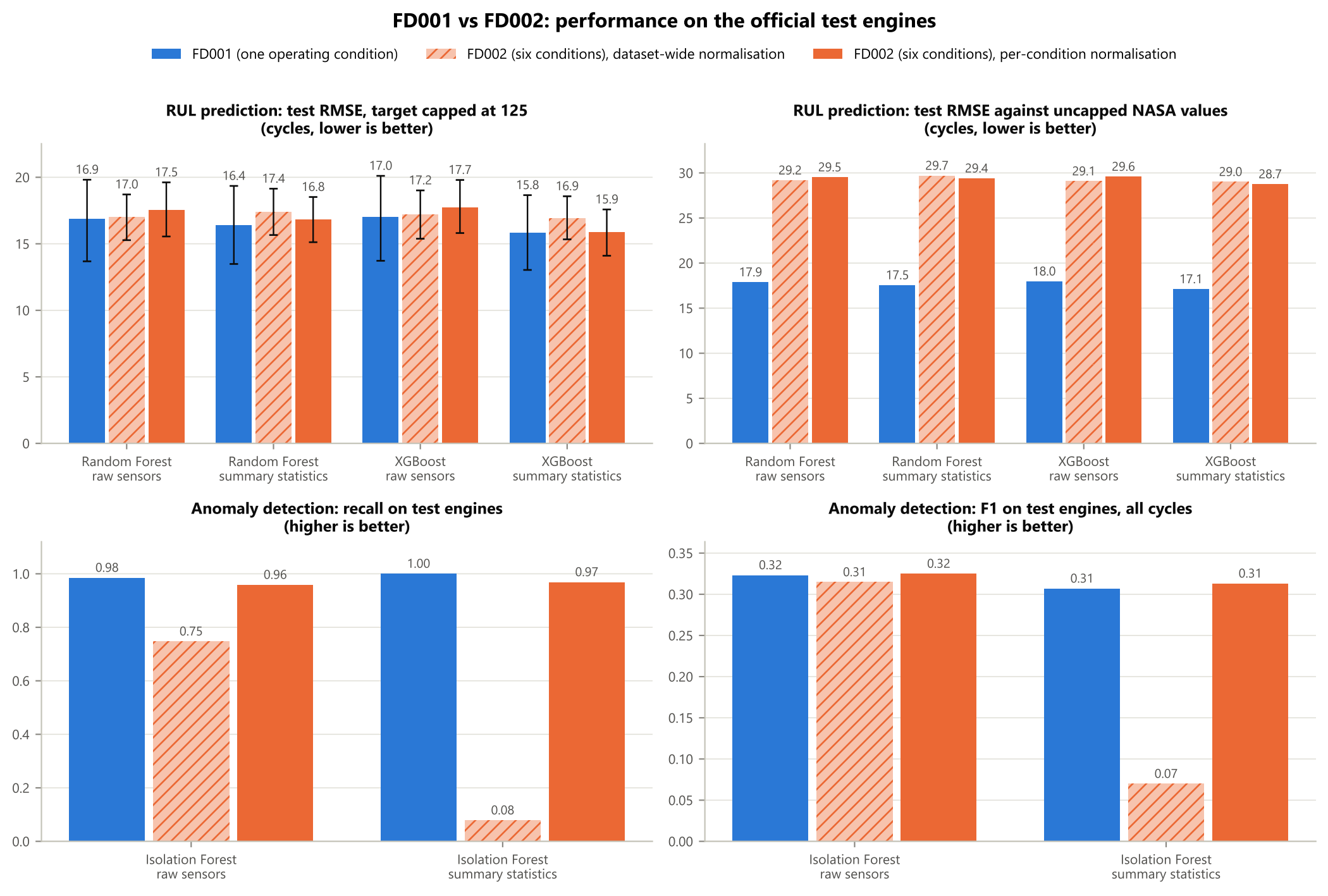
FD001 vs FD002: performance on the official test engines
FD001 (one operating condition)
FD002 (six conditions), dataset-wide normalisation
FD002 (six conditions), per-condition normalisation
RUL prediction: test RMSE, target capped at 125
(cycles, lower is better)
16.9
17.0
17.5
16.4
17.4
16.8
17.0
17.2
17.7
15.8
16.9
15.9
Random Forest
raw sensors
Random Forest
summary statistics
XGBoost
raw sensors
XGBoost
summary statistics
RUL prediction: test RMSE against uncapped NASA values
(cycles, lower is better)
17.9
29.2
29.5
17.5
29.7
29.4
18.0
29.1
29.6
17.1
29.0
28.7
Anomaly detection: recall on test engines
(higher is better)
0.98
0.75
0.96
1.00
0.08
0.97
Isolation Forest
raw sensors
Isolation Forest
summary statistics
Anomaly detection: F1 on test engines, all cycles
(higher is better)
0.32
0.31
0.32
0.31
0.07
0.31
Error bars (top left): 95% bootstrap interval over test engines. Anomaly detection at the 5% alarm budget.